

ASTRE - Lab Assignment 1

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May 2026

Task 1

- Problem 11: [J, A, F, J, E, F, B, F, A, H]
- Problem 12: [E, J, G, B, I, I, E, G, B, I]
- Problem 13: [C, E, I, J, F, D, H, H, J, G]
- Problem 14: [E, J, N, C, J, D, H, N, A, A]
- Problem 15: [L, B, H, A, B, K, K, B, E, A]
- Problem 17: [O, T, H, R, E, I, R, P, I, E]

Problem	Unique branches	Unique error codes	Triggered error codes
11	302	17	0, 9, 24, 26, 39, 42, 48, 52, 60, 62, 74, 75, 82, 91, 93, 94, 95
12	207	0	-
13	392	20	1, 3, 10, 16, 23, 30, 31, 34, 35, 37, 48, 49, 68, 70, 71, 74, 75, 82, 87, 90
14	236	5	18, 28, 41, 73, 77
15	1163	0	-
17	1449	28	0, 2, 11, 12, 19, 22, 23, 25, 35, 37, 46, 50, 53, 56, 61, 63, 66, 69, 72, 76, 78, 79, 85, 86, 95, 96, 97, 99

Table 1: SimpleFuzzer results

Task 2

The results of running the Hill Climber on the RERS problems for 5 minutes can be seen in Table 2. Compared to the random fuzzer from Task 1, the Hill Climber visited more unique branches on problems 12, 14, and 17, while performing worse on problems 11, 13, and 15. The largest improvement was on problem 17, where the Hill Climber visited 1934 branches compared to 1449 for the SimpleFuzzer. The largest regression was on problem 15, where the Hill Climber only visited 757 branches compared to 1163 for the SimpleFuzzer.

Problem	Unique branches	Unique error codes	Triggered error codes
11	284	15	0, 9, 24, 26, 48, 52, 60, 62, 74, 75, 82, 91, 93, 94, 95
12	223	0	-
13	362	18	1, 3, 10, 16, 23, 31, 34, 35, 37, 48, 49, 68, 70, 71, 74, 75, 87, 90
14	253	4	18, 28, 41, 77
15	757	0	-
17	1934	30	0, 2, 11, 12, 19, 22, 23, 25, 31, 35, 37, 46, 50, 51, 53, 56, 61, 63, 66, 69, 72, 76, 78, 79, 85, 86, 95, 96, 97, 99

Table 2: HillClimber results

In general, SimpleFuzzer triggered more unique error codes in most problems. The Hill Climber only outperformed it on problem 17, finding 2 additional error codes (51 and 31). This is likely because the Hill Climber gets stuck in local optima, spending its budget on narrow mutations around a single trace rather than broadly exploring the input space like the SimpleFuzzer does. Increasing `nrMutations` and `MAX_TRACE_LENGTH` to escape local optima more effectively caused timeouts during experimentation, limiting further tuning.

Task 3

The results of running AFL for 5 minutes on each of the RERS problems can be found in Table 3. Compared to the Hill Climber results, we see that AFL finds more unique error codes for problems 12, 13, 14 and 15. It is also able to find error codes for problems which Hill Climber is unable to find any, namely problems 12 and 15. However, Hill Climber finds more unique error codes for problems 11 and 17. Looking at the traces that triggered errors found by both Hill Climber and AFL, we observe that the input sequences differ significantly, indicating that the two approaches explore the space in different ways.

Problem	Unique error codes	Triggered error codes
11	8	0, 48, 60, 62, 74, 82, 91, 95
12	12	3, 10, 25, 26, 37, 65, 71, 73, 74, 76, 77, 79
13	20	1, 3, 10, 16, 23, 30, 34, 35, 37, 48, 49, 52, 68, 70, 71, 74, 75, 82, 87, 90
14	10	18, 28, 36, 41, 59, 60, 73, 76, 77, 90
15	18	4, 7, 12, 21, 25, 27, 28, 33, 35, 44, 71, 72, 74, 76, 78, 84, 87, 93
17	21	2, 11, 19, 22, 23, 25, 31, 46, 50, 51, 53, 61, 63, 66, 69, 76, 79, 95, 96, 97, 99

Table 3: AFL results

References

- [1] Source code repository: <https://github.com/AmyAmyy/ASTRE26/tree/Lab1>